Tugas 2

Nama (NIM) : jordy anugrah akbar (123140141)

Kelas : Sistem Informasi Geografis

Link Github : https://github.com/jordyakbar1/

Deskripsi Tugas :

Lanjutkan database dari Praktikum 1. Tambahkan tabel untuk menyimpan data jalan (LineString) dan wilayah kelurahan (Polygon) di sekitar tempat tinggal Anda. Lakukan konversi format dan validasi data.

Ketentuan Tugas:

- Buat tabel jalan dengan minimal 3 data LineString
- Buat tabel wilayah dengan minimal 2 data Polygon
- Screenshot hasil ST_AsText(), ST_AsGeoJSON(), dan ST_IsValid()
- Tampilkan semua data di QGIS dengan layer berbeda

Langkah-langkah :

1. Buat table “jalan”

Query  Query History

```
CREATE TABLE jalan (
    id SERIAL PRIMARY KEY,
    nama VARCHAR(100),
    geom GEOMETRY(LineString, 4326)
);
```

2. Insert data jalan beserta koordinat (Longitude,Latitude) yang dicopy dari gmaps

Query | Query History

```sql
INSERT INTO jalan (nama, geom)
VALUES

('Gg Alun Alun 3',
ST_SetSRID(
ST_MakeLine(ARRAY[
ST_MakePoint(105.22490455201968, -5.370878627016326),
ST_MakePoint(105.22420264106734, -5.37092453522195),
ST_MakePoint(105.2236178637234, -5.370940519671922)
]),4326)
),

('Gg Alun Alun 2',
ST_SetSRID(
ST_MakeLine(ARRAY[
ST_MakePoint(105.22364823800815, -5.371261426171925),
ST_MakePoint(105.22450195253543, -5.371207298453109),
ST_MakePoint(105.22503859733882, -5.371167847939521)
]),4326)
),

('Gg Alun Alun 1',
ST_SetSRID(
ST_MakeLine(ARRAY[
ST_MakePoint(105.2250253910857, -5.371470052214283),
ST_MakePoint(105.22443881996594, -5.37152985200247),
ST_MakePoint(105.22367495974429, -5.371568220178427)
]),4326)
);
```

ST_AsText(geom)

Query  Query History  Scratch Pad

```
1  SELECT nama, ST_AsText(geom)
2  FROM jalan;
3
4
```

Data Output  Messages  Notifications

Showing rows: 1 to 3  Page No: 1 of 1

| | nama character varying (100) | st_astext text |
|---|---|---|
| 1 | Gg Alun Alun 3 | LINESTRING(105.22490455201968 -5.370878627016326,105.22420264106734 -5.3709245352219 |
| 2 | Gg Alun Alun 2 | LINESTRING(105.22364823800815 -5.371261426171925,105.22450195253543 -5.3712072984531 |
| 3 | Gg Alun Alun 1 | LINESTRING(105.2250253910857 -5.371470052214283,105.22443881996594 -5.3715298520024 |

ST_AsGeoJSON(

Query Query History Scratch Pad

```
1  SELECT nama, ST_AsGeoJSON(geom)
2  FROM jalan;
3
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

| | nama character varying (100) | st_asgeojson text |
|---|---|---|
| 1 | Gg Alun Alun 3 | {"type":"LineString","coordinates":[[105.224904552,-5.370878627],[105.2242026 |
| 2 | Gg Alun Alun 2 | {"type":"LineString","coordinates":[[105.223648238,-5.371261426],[105.2245019 |
| 3 | Gg Alun Alun 1 | {"type":"LineString","coordinates":[[105.225025391,-5.371470052],[105.2244388 |

ST_IsValid()

Query Query History

```
SELECT nama, ST_IsValid(geom)
FROM jalan;
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

| | nama character varying (100) | st_isvalid boolean |
|---|---|---|
| 1 | Gg Alun Alun 3 | true |
| 2 | Gg Alun Alun 2 | true |
| 3 | Gg Alun Alun 1 | true |

Buat table wilayah

Query  Query History

```sql
CREATE TABLE wilayah (
    id SERIAL PRIMARY KEY,
    nama VARCHAR(100),
    geom GEOMETRY(Polygon, 4326)
);
```

Data Output  Messages  Notifications

```
CREATE TABLE

Query returned successfully in 84 msec.
```

Insert 2 Data Polygon

No limit

Query | Query History

```
('Wilayah Rajabasa Timur',
ST_SetSRID(
ST_MakePolygon(
ST_MakeLine(ARRAY[
ST_MakePoint(105.225500, -5.370500),
ST_MakePoint(105.227500, -5.370500),
ST_MakePoint(105.227500, -5.372500),
ST_MakePoint(105.225500, -5.372500),
ST_MakePoint(105.225500, -5.370500)
])
),4326)
);
```

Data Output | Messages | Notifications

```
INSERT 0 2

Query returned successfully in 68 msec.
```

Query  Query History

```
SELECT nama, ST_AsText(geom)
FROM wilayah;
```

Data Output  Messages  Notifications

Showing rows: 1 to 2  Page No: 1 of 1

| | nama character varying (100) | st_astext text |
|---|---|---|
| 1 | Wilayah Rajabasa Barat | POLYGON((105.2235 -5.3705,105.2255 -5.3705,10 |
| 2 | Wilayah Rajabasa Timur | POLYGON((105.2255 -5.3705,105.2275 -5.3705,10 |

Query  Query History  Scratch Pad

```
SELECT nama, ST_AsGeoJSON(geom)
FROM wilayah;
```

Data Output  Messages  Notifications

Showing rows: 1 to 2  Page No: 1 of 1

| | nama character varying (100) | st_asgeojson text |
|---|---|---|
| 1 | Wilayah Rajabasa Barat | {"type":"Polygon","coordinates":[[[105.2235,-5.3705],[105.2255,-5.3705],[105.2255,-5.3725],[105.22 |
| 2 | Wilayah Rajabasa Timur | {"type":"Polygon","coordinates":[[[105.2255,-5.3705],[105.2275,-5.3705],[105.2275,-5.3725],[105.22 |

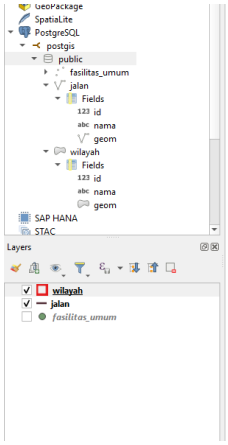
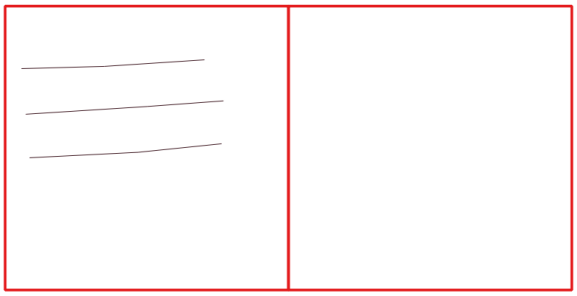
No limit

Query | Query History

```
SELECT nama, ST_IsValid(geom)
FROM wilayah;
```

Data Output | Messages | Notifications

Showing rows: 1 to 2 | Page No: 1 of 1

| | nama character varying (100) | st_isvalid boolean |
|---|---|---|
| 1 | Wilayah Rajabasa Barat | true |
| 2 | Wilayah Rajabasa Timur | true |

Hasil QGIS